

BCSG 0802: COMPUTER PROGRAMMING-I

Objective: To impart adequate knowledge on the need of problem-solving techniques and develop programming skills to implements applications using the concepts of C Language. Also, by learning the programming constructs they can easily switch over to any other language in future.

Credits:04
0-8-0

L-T-P-J: 0-

Module No.	Content	Teaching Hours
I	Generation of Programming Languages: Low, Assembly, High and 4GL. Language Processors: Compiler, Interpreter, Assembler, Linker and Loader. Algorithm: Introduction, Features, Different Ways of stating Algorithms. Flow Chart: Introduction, Standard, Guidelines, Advantages and Limitations of using Flowcharts. Basics of C: Overview, Structure of a C program, Identifier, Keywords, Variables, Data types, Formatted Input and output. Operators and Expression: Assignment, Unary, Arithmetic, Relational, Logical, Bitwise, Conditional, Special operators and their precedence & Associativity. IEEE representation of data types like float & double, Lvalue and Rvalue Type Conversion: Type Promotion in expression, Conversion by Assignment, Truncation and Casting Arithmetic expression. Decision and Case Control Structure: if, if-else, nested if-else, Decisions using switch, switch versus if-else ladder, goto. Loop Control Structure: For loop, while loop, do-while loop, nesting of loops, break, and continue. Arrays: Introduction, one-dimensional and two-dimensional Array-Declaration, Initialization, Address Calculation. Operations on Arrays: Insertion, Deletion, Linear Search & Bubble Sort. String: Introduction, One dimensional and two-dimensional Array-Declarations, Initialization Operations on String: Length, Copy, Reverse, Concatenate, Compare with & without built-in functions.	25
II	Functions: Declaration and Definition, Category of Functions, Parameter Passing Techniques – Call by Value, Passing Arrays to Functions. Introduction to Storage Classes: Auto, Static, Extern and Register. Recursion: Mechanics of Recursive Call, Implementation of Recursion, Recursion vs. Iteration. The C Preprocessor: Introduction, Macro Expansion and File Inclusion, Conditional Compilation and Miscellaneous Directives. Pointers: Declaration and Initialization of Pointer Variables, Accessing a Variable through its Pointer, Arrays and Pointers, Pointer and Strings, Pointer Arithmetic, Pointers to Pointers, Array of Pointers, Pointer to an Array, Two Dimensional Array and Pointers, Pointers to Functions, Dynamic Memory Allocation, void Pointer and Null Pointer. User Defined Types: enum, typedef, Union and Structure - Declaration, Initialization, Nested Structures, Arrays of Structures, Structure and Pointer, Passing Structure Through Function. Difference between Structures and Union. File Handling: Data and Information, File Concepts, File Organization, File Operations: Open, Read, and Close, Trouble in Opening a File. File Opening Modes, Working with Text Files. Random Access to Files of Records. Introduction to Command Line Arguments.	25

Text Books:

- Behrouz A. Forouzan and Richard F. Gilberg, "Computer Science – A Structured Programming Approach Using C", C Language Learning, 2007

Reference Books:

- Herbert Schildt, "C: The Complete Reference", 5th Edition, McGraw Hill Education
- K. N. King, "C Programming a Modern Approach", W. W. Norton, 2nd Edition, 2008.
- Kernighan and Ritchie, "The C Programming Language", PHI, 2nd Edition, 2011.
- P. Dey and M. Ghosh, "Programming in C", Oxford University Press 2nd Edition, 2013.

Focus: This Course focuses on Employability under CO2, CO3, CO7.

Outcome: After completion of course, the student will be able to:

- CO1: Understand the basic concepts of problem-solving skills.
- CO2: Apply the basic principles of programming in C language.
- CO3: Understand the concepts of arrays and strings in C language.
- CO4: Apply the concepts of functions to solve real world problems.
- CO5: Illustrate the concepts of recursion.
- CO6: Understand the concepts of pointers in C language.
- CO7: Understand the basic concepts of file handling.
- CO8: Develop algorithmic solutions to simple computational problems.

Mapping of Course Outcomes (COs) with Program Outcomes (POs) and Program Specific Outcomes (PSOs):

COs	POs/PSOs
CO1	PO1,PO2,PO4,PO12/PS01,PS03
CO2	PO1,PO2,PO3,PO10/PS01,PS03
CO3	PO1,PO2,PO3,PO4/PS01,PS03
CO4	PO1,PO3, PO12/PS01,PS02
CO5	PO1,PO2,PO4 /PS01,PS03
CO6	PO1,PO2,PO3,PO4/PS01,PS02
CO7	PO1,PO3,PO6 /PS01
CO8	PO1,PO2,PO4,PO10,PO12/PS01,PS03